

# Representation Convergence - Status at a Glance

All models pretrained and frozen; only closed-form linear maps fitted, except rows marked (trained). Thesis: independently trained models share representations up to a linear transformation - and that convergence is deployable.

| Exp      | Models                                     | Input data (modality; train / eval split)                                                                                                                                                                                                                                      | Target                                                          | Achievement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | GPT-2 vs Pythia-160M                       | <b>TEXT only</b> - 10k WikiText passages, no labels or captions. Fit 7,500 (9.8 rows/dim) / <b>eval 2,500</b> held out, 75:25 split                                                                                                                                            | Do two independently trained LLMs converge?                     | Yes, linearly: stitching R2 0.74-0.80 vs 0.17 random control; CKA 5.3x control; correspondence real but hub-dominated; eff. dim 2.2-10.9 vs 9.3-92.7 of 768 explains the metric artifacts                                                                                                                                                                                                                                                                                                     |
| B        | MobileCLIP-S1 -> SigLIP 2                  | <b>IMAGES only</b> for the fit - 4k COCO val2017, each encoded twice; captions enter only at evaluation. Fit 3,000 (5.9 rows/dim, bottom of band) / <b>eval 1,000</b>                                                                                                          | Are two vision encoders linearly interchangeable?               | Ridge R2 0.592 / cosine 0.900; Procrustes fails (0.038) -> anisotropy (69x spread in W's singular values); MLP +0.003 -> the relationship is exactly linear                                                                                                                                                                                                                                                                                                                                   |
| B3/D     | same + Qdrant index                        | <b>IMAGE + TEXT</b> - no fitting. <b>Eval 1,000</b> : the B held-out images as the gallery, their first captions as queries (gallery size 1,000 throughout)                                                                                                                    | Is the convergence deployable?                                  | One 0.79 MB fp16 adapter, one shared index: R@5 = 95.1% of ceiling; raw = chance; free extra capability: cross-tier membership R@1 0.987                                                                                                                                                                                                                                                                                                                                                      |
| E1       | DINOv2 -> bge-m3                           | <b>IMAGE-CAPTION pairs</b> - 8k COCO train2017. All <b>5</b> captions (what COCO provides) averaged per image: cancels annotator idiosyncrasy, R2 0.278 -> 0.511. Fit 7,000 (9.1 rows/dim) / <b>eval 1,000</b> ; ceiling re-measured under the same averaging (0.648 -> 0.875) | Does convergence cross modalities without joint training? (PRH) | Measurable but weak: R2 0.511; R@1 0.358 = 40.9% of a matched SigLIP ceiling, 358x chance; verdict robust across caption protocols; cross-modal costs ~half of same-modality R@1                                                                                                                                                                                                                                                                                                              |
| E1 sweep | DINOv2 small / base / large -> bge-m3      | <b>IMAGE-CAPTION pairs</b> , same protocol and cls+patch pooling at every point. 9.5k / 18k / 23.8k pairs - N scaled with output width so rows/dim is <b>11.1</b> throughout; <b>eval 1,000</b> each, ceiling re-measured per run                                              | Does alignment grow with encoder capacity? (the PRH trend)      | <b>Yes - measured.</b> R@1 = <b>40.9 / 53.5 / 58.3%</b> of a matched ceiling across 13.8x params; monotone in R2 (0.516/0.580/0.609), cosine, all three recalls AND the shuffle gap (0.616/0.682/0.709). +15.3 points of ceiling per decade (fit R2 0.949); base and large exceed the same-modality 0.592. Returns are compressive - no extrapolation beyond 304M                                                                                                                             |
| E1.1     | DINOv2 -> GPT-2 (vs bge-m3, matched pair)  | <b>IMAGE-CAPTION pairs</b> , identical to E1 by construction. Fit 7,000 (9.1 rows/dim) / <b>eval 1,000</b>                                                                                                                                                                     | Does the text side need similarity training?                    | Predicted split: R2 retains 74%, retrieval 27%; GPT-2 space is near-1D (eff. rank 6.5/768, pair-cos +0.999) so cosine is degenerate (reads 1.000 at 9.5% retrieval) - metric validity depends on the space's geometry                                                                                                                                                                                                                                                                         |
| E1.2     | DINOv2 -> GPT-2 + label-free corrections   | <b>no new data</b> - reuses the E1.1 embeddings (base, cls+patch). Transforms fitted on the 17,066 train targets only; <b>eval 1,000</b>                                                                                                                                       | Is the collapse missing correspondence, or missing isotropy?    | <b>Isotropy.</b> Whitening - unsupervised, label-free - lifts R@1 0.130 -> <b>0.478</b> , reaching <b>102.6%</b> of the contrastively trained bge-m3 arm. Effective rank vs R@1 tracks monotonically (rho +0.94). Centering alone fails (fixes pair-cosine, not spread). Whitened R2 is <b>0.014</b> with the best retrieval of any arm - the metrics fully decouple                                                                                                                          |
| E2/E3    | the E1 stack + SigLIP text tower           | E2 demo: <b>500 unseen val2017</b> images + captions as the gallery. E3: <b>eval 1,000</b> (the B3 rows), plus a 200-3,000 row data-efficiency curve                                                                                                                           | Demo both search directions; can a consumer USE the map?        | Working two-way demo; stitching penalty R@5 = 0.049 -> the unchanged SigLIP text tower consumes translated vectors near-transparently                                                                                                                                                                                                                                                                                                                                                         |
| F1-F2    | SigLIP 2 teacher, MLP head (trained)       | <b>IMAGES only, NO captions</b> - the teacher supplies the target. Fit 38,000 (74 rows/dim; MLP 0.029 examples/param) / <b>eval 2,000</b> in-corpus, plus the 1,000 B3 rows for comparable retrieval                                                                           | Was the +0.003 MLP control data-starved?                        | No: at 74 rows/input-dim the MLP converges to EXACTLY the linear solution (0.9052 = 0.9052) - bottleneck confirmed twice                                                                                                                                                                                                                                                                                                                                                                      |
| C1.1     | MobileCLIP <-> SigLIP 2, no pairing        | <b>IMAGES only</b> , 1,500 vectors per side. The pairing is <b>destroyed</b> and the true permutation sealed until scoring; eval on the standard 1,000 rows                                                                                                                    | Can the correspondence be found with NO paired examples?        | <b>Partially, yes.</b> Gromov-Wasserstein on shape alone recovers <b>39.7%</b> exact matches ( <b>595x</b> chance); the map fitted from it reaches R@1 0.252 = 40% of ceiling, verification AUC 0.906 - with zero pairs. Two measured negatives: ICP degrades it, and whitening (which rescues retrieval in E1.2) destroys it - isotropy helps you READ a correspondence, hurts you when you must FIND one                                                                                    |
| G1       | 7 encoders -> one 512-d hub                | <b>7 cached spaces</b> , same 9,533 images: DINOv2 small/base/large, bge-m3, GPT-2, BERT, SBERT - <b>4 text objectives, 2 modalities</b> . Balanced whitened-PCA hub; one linear map each; 8,533 train / <b>eval 1,000</b> ; width x alpha swept                               | Is there ONE shared coordinate system, usable across encoders?  | <b>Yes.</b> Head trained on ONE encoder reaches <b>95.9%/92.9%</b> of native on encoders it NEVER saw; 7-space transfer R@1 <b>0.451</b> (AUC 0.986) - UNCHANGED from 5-space (0.445; diff within noise). Composition retains <b>95.6%</b> of pairwise maps on 36/41 pairs. One exception, measured not assumed: writing INTO GPT-2 (24.8%) - reading OUT works (95.1%). Shape agreement before any map: cross-modal rho +0.29; bge-SBERT +0.76 highest of all, sharing objective not lineage |
| E1.3     | encoder ladder: GPT-2 / BERT / SBERT / bge | <b>same image cache</b> (DINOv2-base cls+patch, 18k pairs, 11.1 rows/dim); only the TEXT objective varies. Predictions <b>pre-registered</b> before running                                                                                                                    | Does C.11's mechanism predict encoders it never saw?            | <b>P1 confirmed exactly:</b> raw R@1 orders by eff.rank, rank-corr + <b>1.00</b> (6.5/65/93 -> 0.160/0.362/0.510); BERT lands between GPT-2 and SBERT as registered. <b>P2 partial - the sharper finding:</b> whitening lifts every arm (0.478/0.521/0.588) but a 0.110 spread remains in raw order: isotropy is <b>necessary, not sufficient</b> - contrastive training also arranges WHICH directions carry meaning, and whitening cannot install that                                      |
| D.2      | adapter as a threshold decoder             | <b>1,000 held-out images</b> , the shipped adapter scored as three decision rules (verification / ranking / membership)                                                                                                                                                        | How well does one map decide, not just rank?                    | <b>A hierarchy.</b> Verification AUC <b>0.999</b> , per-query ranking R@1 <b>0.987</b> , membership under one global cutoff AUC <b>0.889</b> (best F1 0.840). Harder as the decision gets more global (best-of-500 impostor). ROC fits thresholded tasks, not R@K; AUC measures separation not magnitude (caption-verif AUC 0.997 at cosine 0.136)                                                                                                                                            |
| F5       | refit linear W (38k rows)                  | Fit <b>images only</b> (38,000). <b>Eval 5,000</b> image-caption pairs from train2017 beyond the corpus - 5 disjoint 1,000-galleries, unseen by both maps (5x the usual eval, to settle R@1)                                                                                   | Is the shipped adapter improvable by data alone?                | Yes, significantly: R@1 +0.024, R@5 +0.012, R@10 +0.011 on 5,000 fresh paired queries (all CIs exclude 0); same artifact, same seconds-long refit; 90%-of-ceiling gate approached (89.0%), not met                                                                                                                                                                                                                                                                                            |
| F3-F4    | MobileCLIP last stage (trainable)          | <b>IMAGES only</b> , same 38k corpus restreamed (the trunk trains). Fit 38,000 / <b>eval 2,000</b> , then the 1,000 B3 rows                                                                                                                                                    | Can the ceiling itself move?                                    | Built, deliberately gated: justified by F2's verdict but trades the seconds-refit property for full retraining                                                                                                                                                                                                                                                                                                                                                                                |

**No training in A/B/E:** all models frozen (no\_grad); the only fitted object is a closed-form ridge matrix W outside both models; weights are bit-identical after every experiment. Captions never enter DINOv2; pixels never enter bge-m3 - the pairing is row-index bookkeeping. F-series rows train new artifacts beside the frozen encoders (F2: a separate head); F3, which would touch encoder weights, is built but unrun - every pretrained model remains bit-identical to its download.

**Bottom line:** convergence confirmed on three model pairs (text-text, vision-vision, vision-text); exactly linear; deviation from rigidity is measured anisotropy; shipped as a 0.79 MB adapter at 95% of ceiling and improved further by refitting on more unlabeled data. Sharpest methodological finding: similarity metrics are only interpretable after measuring the space's geometry - cosine read 1.000 in a space where retrieval ran at 9.5%.